RegolithFleet: Comparing Coordination Strategies for Multi-Robot Regolith Excavation and Haulage in a Grid-Based Lunar Outpost

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Future lunar and planetary surface operations are expected to rely on teams of autonomous robots for excavation, site preparation, and regolith transport. Efficient coordination is especially important when multiple robots must repeatedly travel between a shared depot and an expanding excavation frontier, where congestion, route conflicts, and inefficient task allocation can significantly reduce overall mission performance. This work presents RegolithFleet, a ROS 2-based simulation framework for studying multi-robot regolith excavation and haulage in a grid-based lunar outpost scenario.

In the proposed setting, a team of robots starts from a shared depot and must excavate a predefined dig-site blueprint by repeatedly traveling to frontier cells, removing one unit of regolith, and returning the material to the depot. The project investigates how different coordination strategies affect mission efficiency under identical environmental and task conditions. Three approaches are considered: an uncoordinated greedy baseline, a centralized coordination method with global task assignment, and a decentralized coordination strategy based on local decision rules. The comparison focuses on metrics including total completion time, total travel distance, congestion events, rerouting frequency, and robot utilization.

The main objective is to identify how coordination structure influences excavation throughput and congestion as fleet size and site complexity increase. The simulation is intended as a lightweight experimental platform for studying cooperative excavation logistics in constrained planetary environments and for exploring the trade-offs between coordination quality, scalability, and operational robustness. This work contributes an initial benchmark scenario and an experimental framework for evaluating multi-robot excavation strategies relevant to future robotic lunar infrastructure and in-situ resource utilization tasks.